ML18 A fair comparison: does your model beat the plain guess?

Week 6, Trees and fair comparisons | Session 18 of 24 | Machine Learning | New library: sklearn

Today's question. Which tree depth is better, and does it beat the plain guess on the held-back flowers?

Objective. By the end: you can split data into training, validation and held-back parts, choose a model on validation only, and report one held-back result with its limits.

> **WHERE THIS IS USED**
> A football coach tests two formations on matches the team has not practised for. You do the same with two tree depths. Would you pick a formation after one lucky match?

Before this session: ML17 Mistakes, overfitting and fair comparisons. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
stratify	Keep the share of each category the same in every part.	Making sure every group has some of each village.
validation score	A score on held-out rows used to choose between models.	A trial run.
frozen choice	A decision fixed before the held-back set is used.	Sealing your prediction in an envelope before the match.
split sizes	How many rows go to training, validation and test.	How the class is divided for the three jobs.

Block A. A three-way split

First hold back 20% of the flowers as the held-back set, then take a quarter of what remains for validation. stratify keeps each species equally represented in every part.

Predict. Predict the three sizes. They should add up to 150. Write your guess on paper first.

Your guess, written before you run it

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
X, y = load_iris(return_X_y=True)
X_dev, X_test, y_dev, y_test = train_test_split(X, y, test_size=0.2, random_state=7,
    stratify=y)
X_train, X_val, y_train, y_val = train_test_split(X_dev, y_dev, test_size=0.25,
    random_state=7, stratify=y_dev)
```

```
print(len(X_train), len(X_val), len(X_test))
```

Run it, then compare. The cell prints 90, 30 and 30. stratify=y means each species has the same share in every part: 10 of each species in validation and in the held-back set. The held-back set is now set aside and must not be used to make any choice.

Investigate.

1. Why are the three parts 60%, 20% and 20%?
2. What does stratify do that a plain shuffle does not?

Modify. Print how many flowers of species 0 are in the validation part.

Block B. Choose on validation, test once

Compare a tree of depth 1 and one of depth 3 on validation only. Freeze the winner. Then use the held-back set once.

Predict. Predict which depth wins on validation. Write it down before you run. Write your guess on paper first.

Your guess, written before you run it

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.dummy import DummyClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
baseline = DummyClassifier(strategy="most_frequent").fit(X_train, y_train)
val_baseline = accuracy_score(y_val, baseline.predict(X_val))
tree1 = DecisionTreeClassifier(max_depth=1, random_state=7).fit(X_train, y_train)
tree3 = DecisionTreeClassifier(max_depth=3, random_state=7).fit(X_train, y_train)
score1 = accuracy_score(y_val, tree1.predict(X_val))
score3 = accuracy_score(y_val, tree3.predict(X_val))
if score3 > score1:
    chosen_depth = 3
    final_model = tree3
else:
    chosen_depth = 1
    final_model = tree1
final_acc = accuracy_score(y_test, final_model.predict(X_test))
print(val_baseline, score1, score3, chosen_depth, final_acc)
print(confusion_matrix(y_test, final_model.predict(X_test)))
```

Run it, then compare. The baseline scores exactly one third on validation, because each species is a third. The deeper tree usually wins on validation. The choice is made from the validation scores and then frozen. final_acc is the one honest number: it comes from 30 flowers the choice never saw. The report needs the split sizes, the baseline, both validation scores, the chosen depth, the held-back accuracy, the confusion matrix and two limitations.

Investigate.

1. Which scores did you use to choose the depth?
2. What is one limitation of a held-back set of 30 flowers?

Modify. Say what you would do if the held-back score came out lower than the validation score.

Make

Save ML18.ipynb with: the split sizes, the baseline, two validation scores, the chosen depth, the held-back accuracy, the confusion matrix and two limitations. Explain your decision without claiming that the highest score proves one depth is best in general.

Save your work as ML18. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
The final accuracy was calculated before the choice	The held-back set was used while deciding the depth.	Mark that result as validation and use the instructor's alternate holdout for the final.
ValueError: The least populated class in y has only 1 members	stratify was given labels that are too few.	Check that y is the full label list.

Exit ticket

I choose with validation scores, freeze the choice, then report one held-back score with its limits.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.